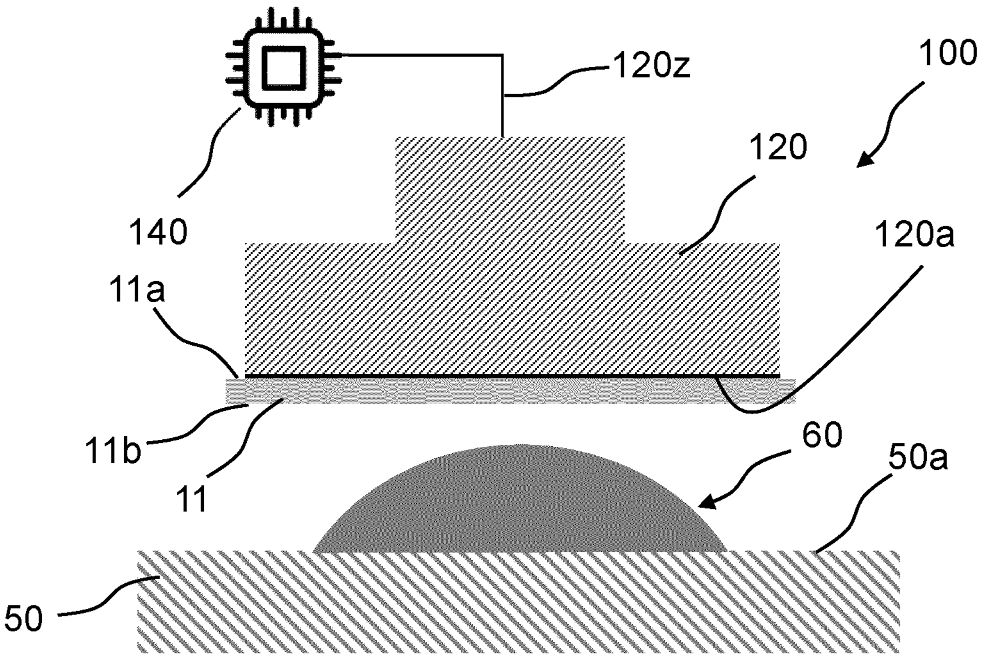
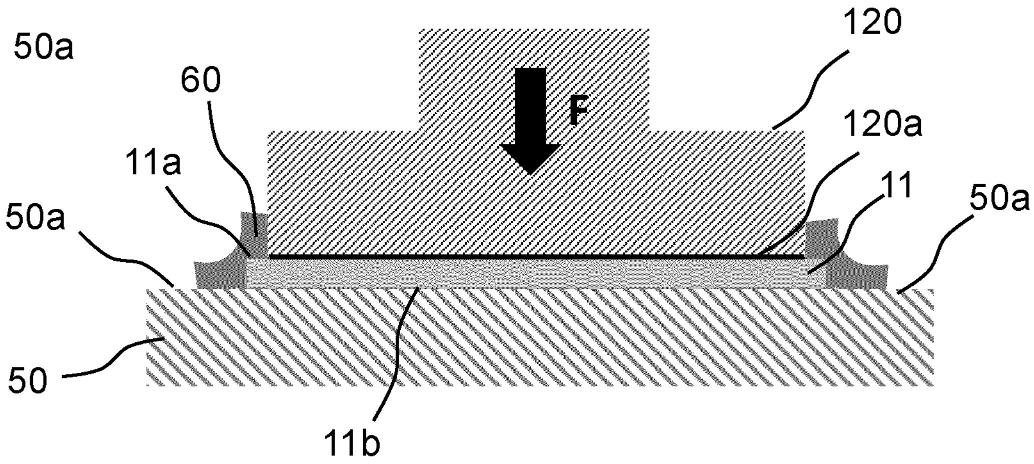


140
120z
100
120
120a
11a
11b
11
60
50a
50

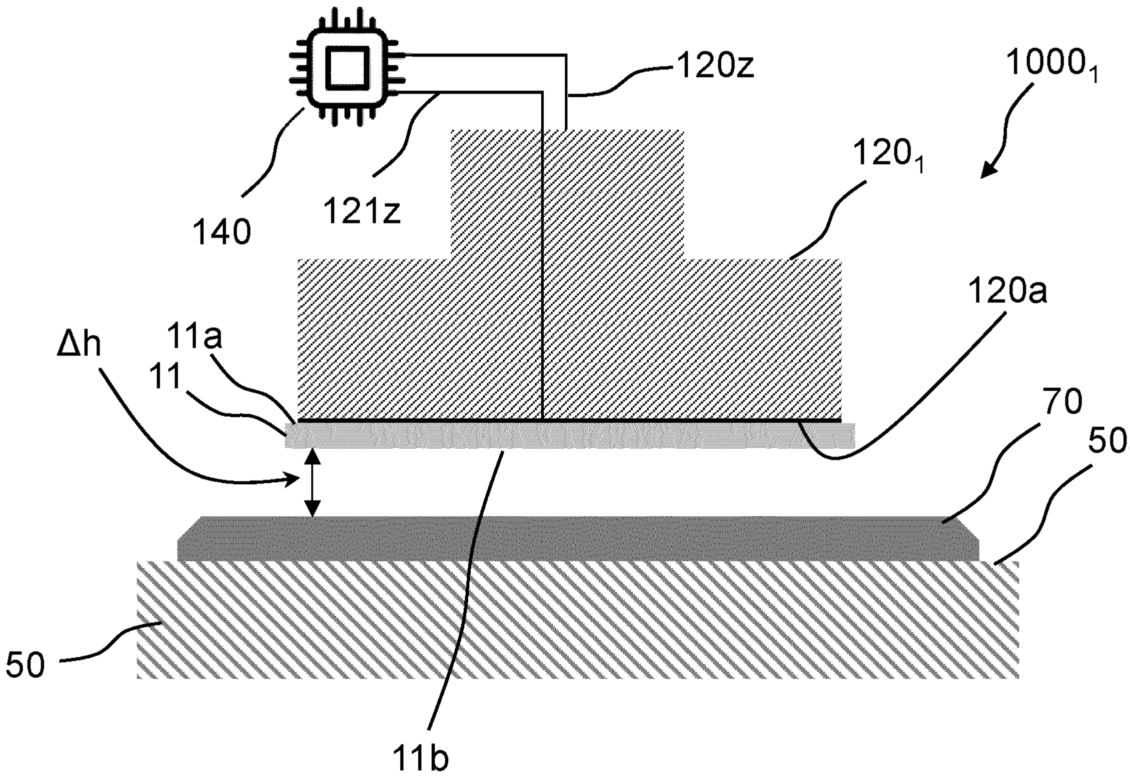
(A)



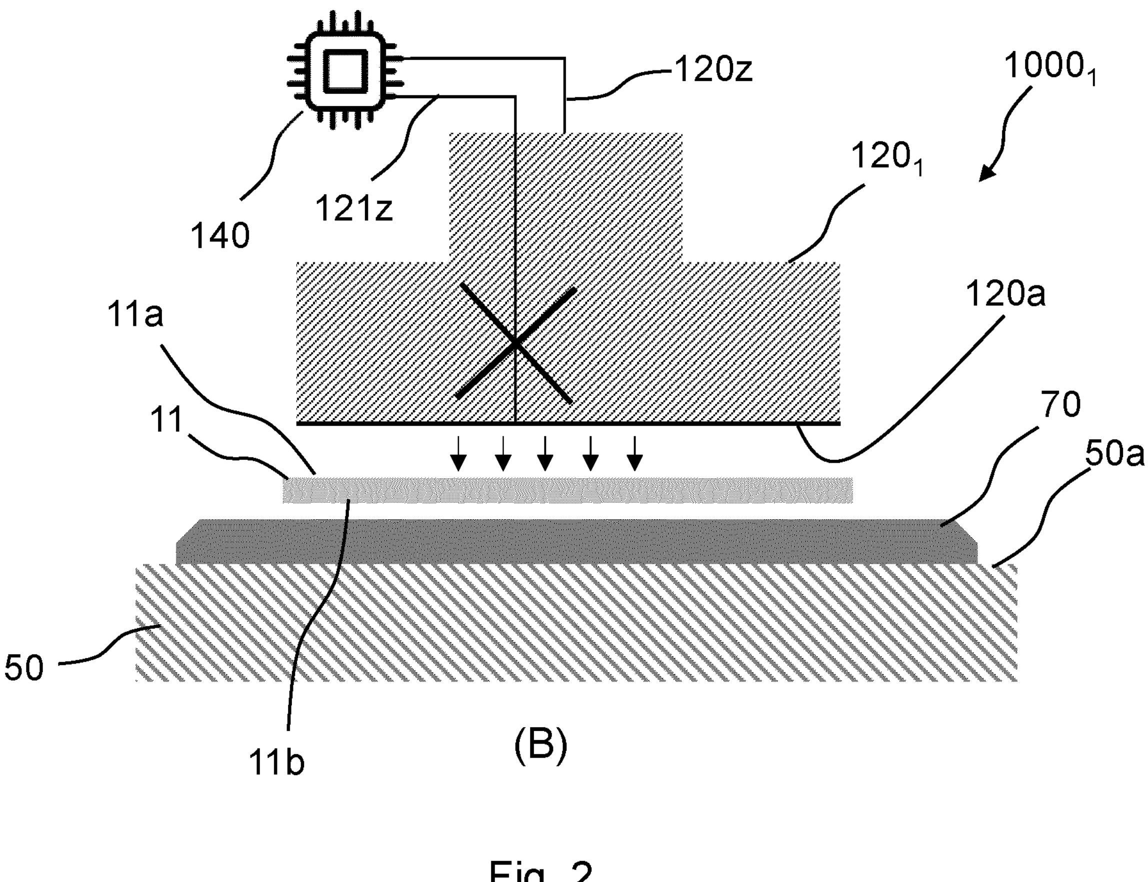
50a
120
F
60
120a
11a
11
50a
50a
50
11b

(B)

Fig. 1

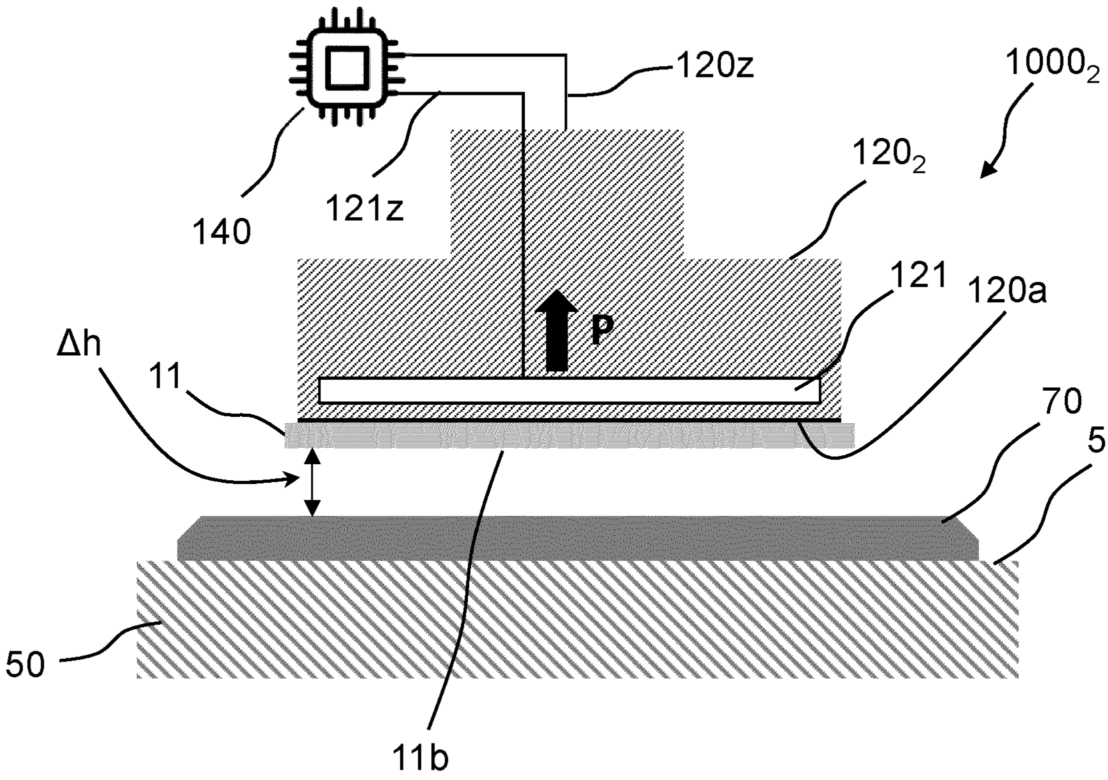

120z
1000_1
140
121z
120_1
120a
Δh
11a
11
70
50a
50
11b

(A)

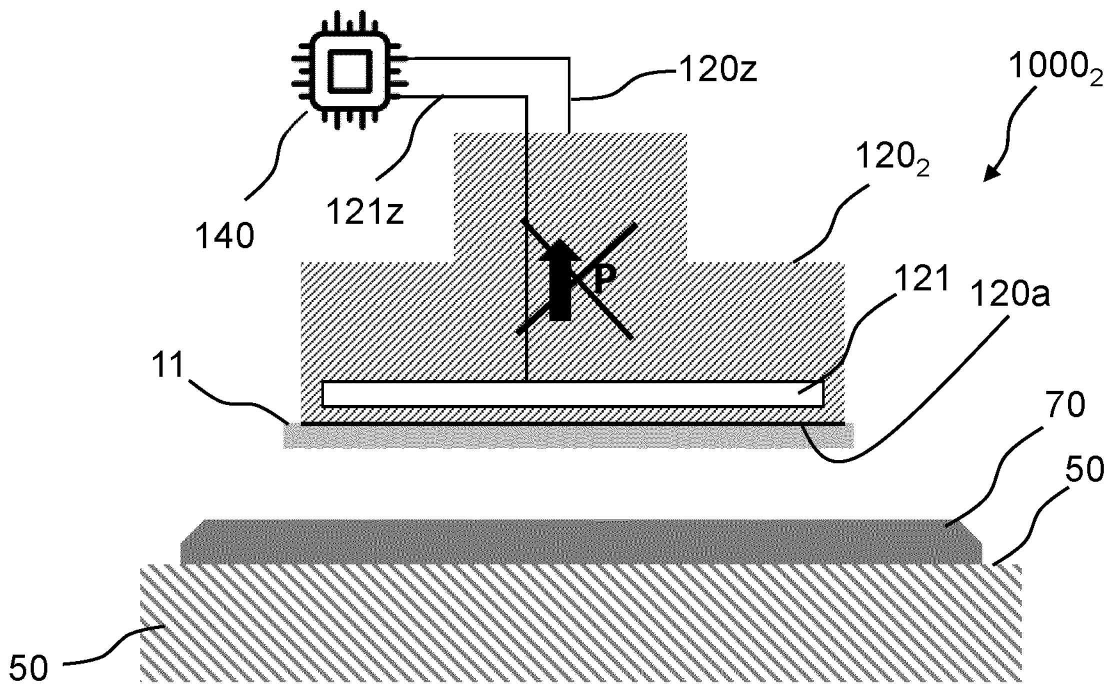

120z
1000_1
140
121z
120_1
120a
11a
11
70
50a
50
11b

(B)

Fig. 2

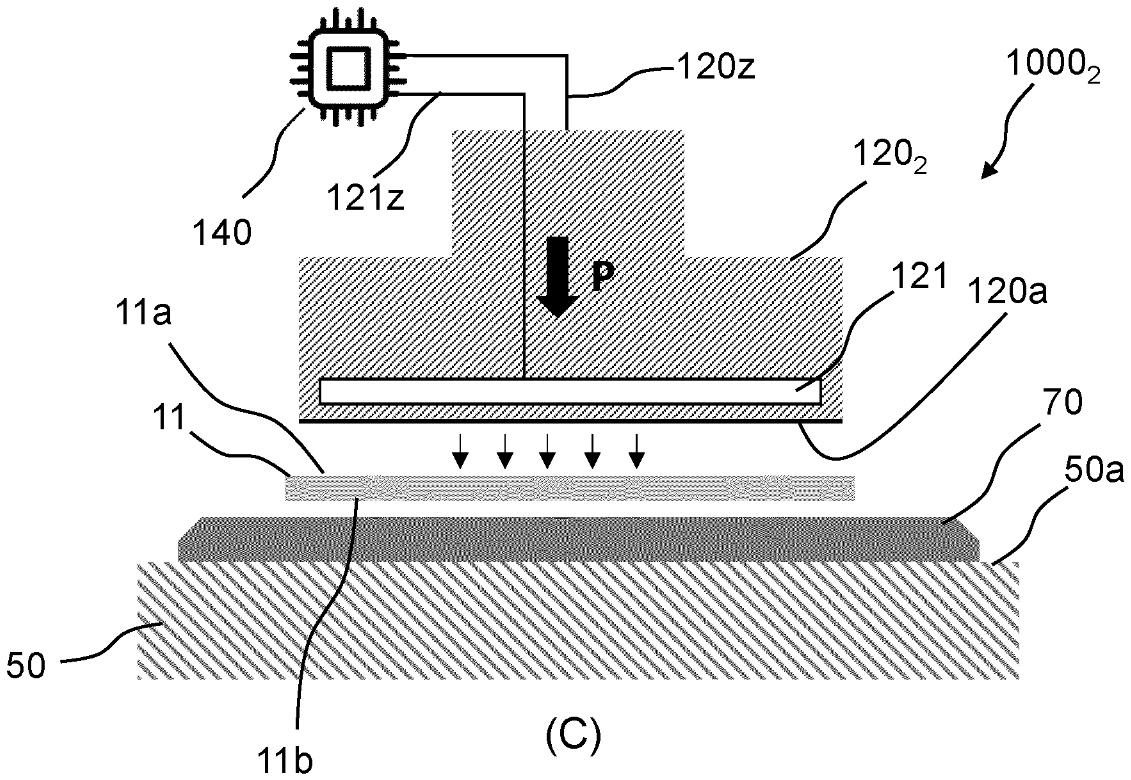

120z
1000_2
140
121z
120_2
121
120a
P
Δh
11
70
50a
50
11b

(A)


120z
1000_2
140
121z
120_2
P
121
120a
11
70
50a
50

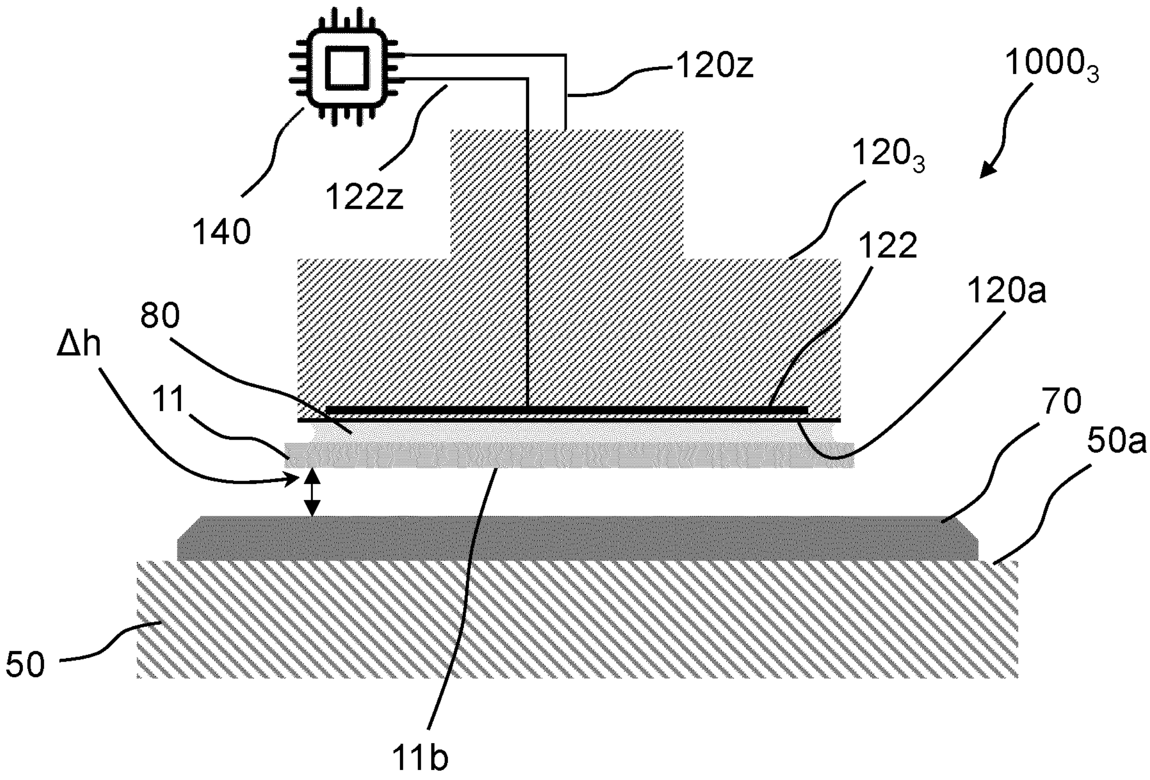
(B)

Fig. 3



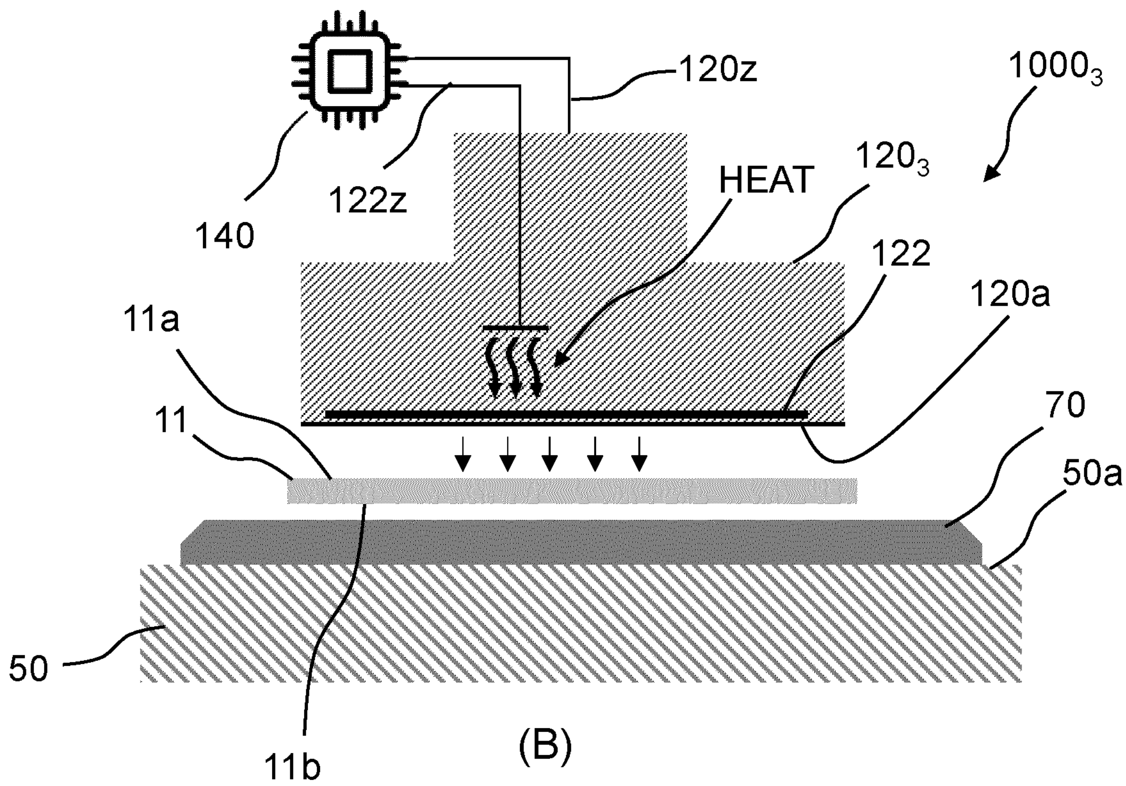
(C)

Fig. 3



120z
1000_3
122z
140
120_3
122
80
120a
Δh
11
70
50a
50
11b

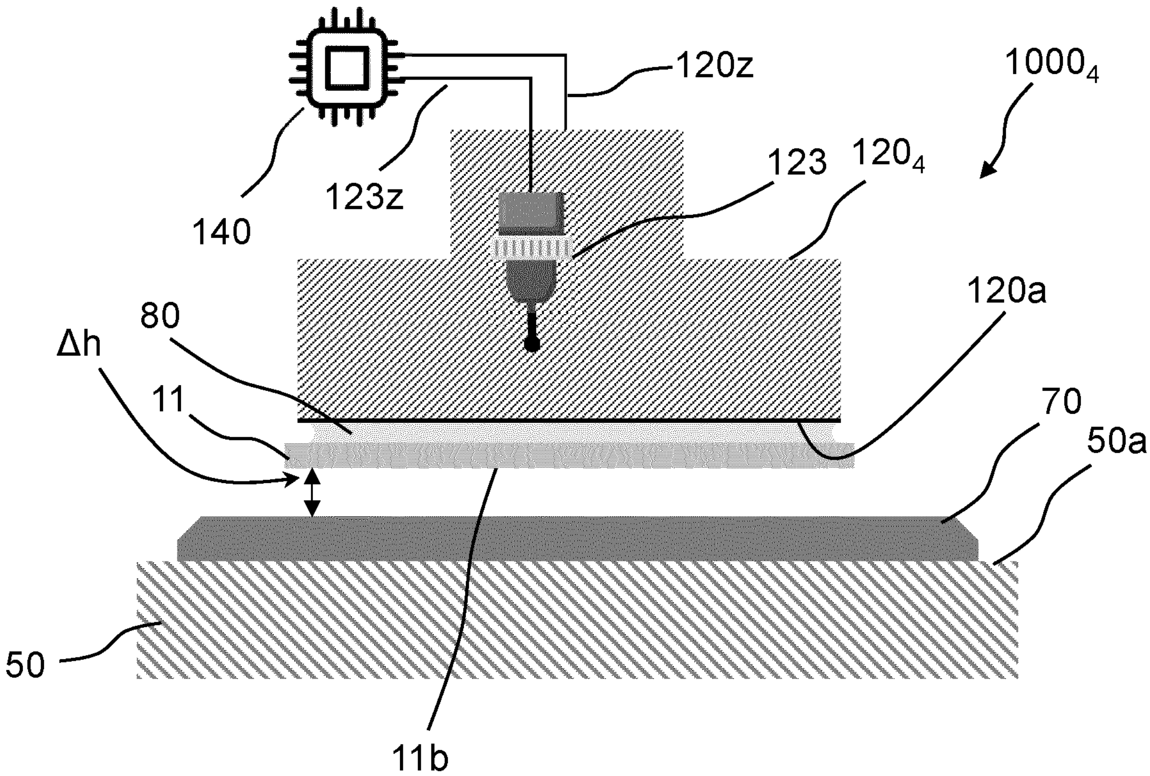
(A)



120z
1000_3
122z
140
HEAT
120_3
122
120a
11a
11
70
50a
50
11b

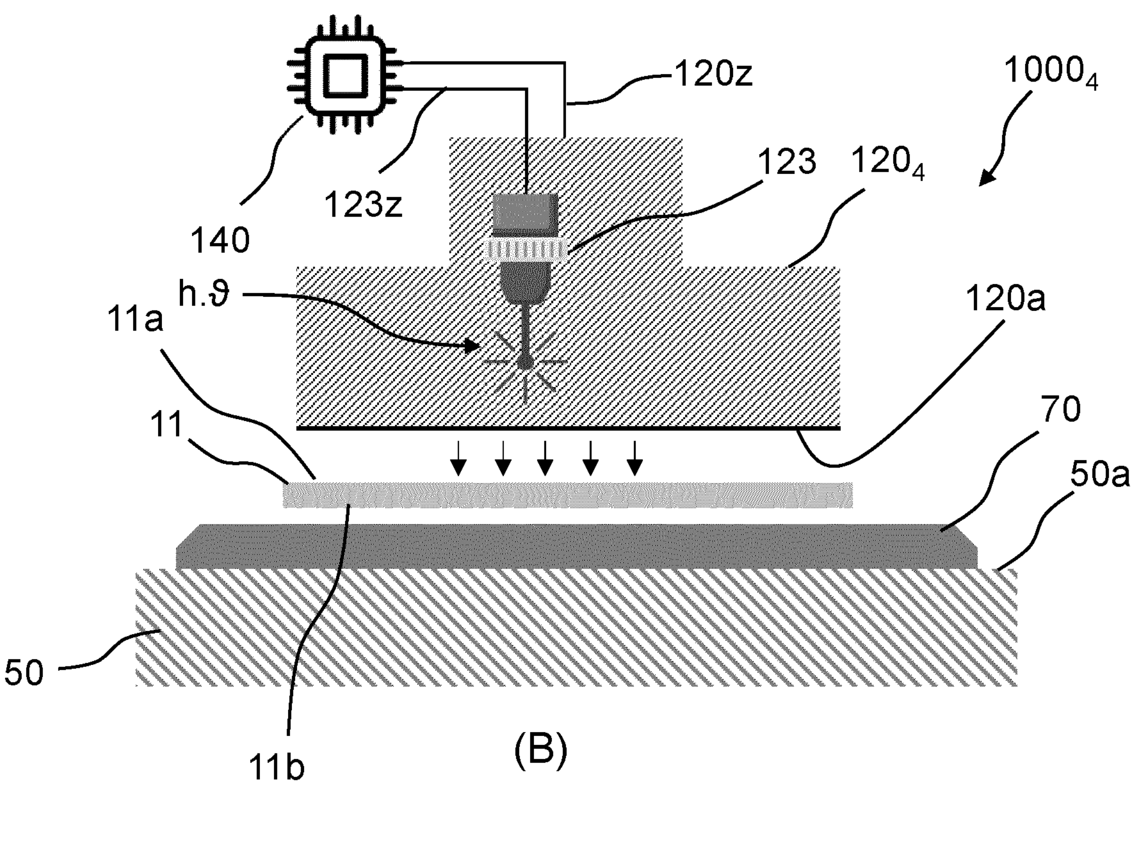
(B)

Fig. 4



120z
1000_4
123z
140
123
120_4
120a
80
Δh
11
70
50a
50
11b

(A)



120z
1000_4
123z
140
123
120_4
h.ϑ
11a
120a
11
70
50a
50
11b

(B)

Fig. 5

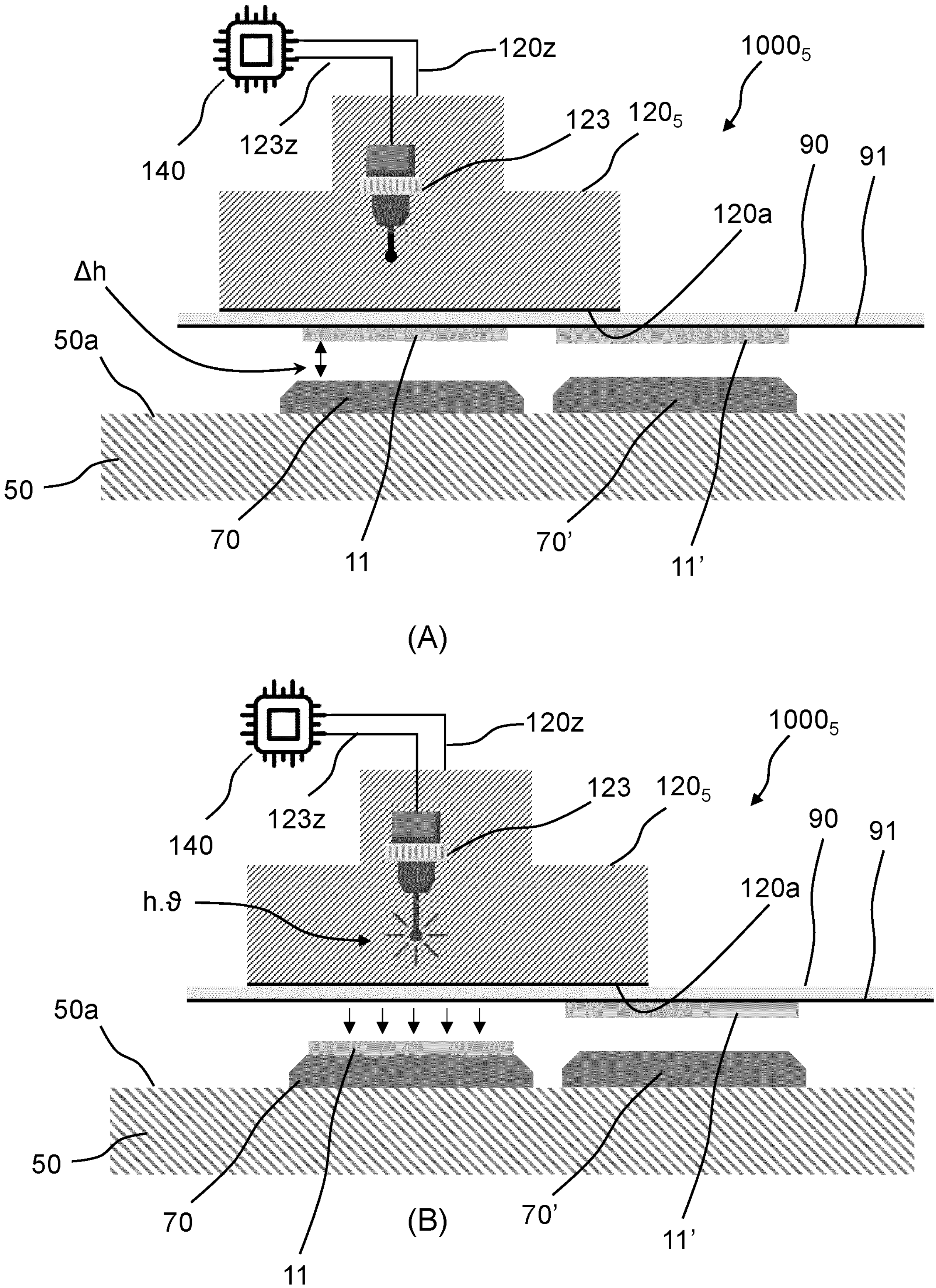

120z
1000_5
140
123z
123
120_5
90
91
120a
Δh
50a
50
70
11
70'
11'
(A)
120z
1000_5
140
123z
123
120_5
90
91
h.ϑ
120a
50a
50
70
11
70'
11'
(B)

Fig. 6

TITLE

A method for mounting an electronics component at a particular placement location on a carrier as well as a pick-and-place apparatus for mounting an electronics component at a particular placement location on a carrier.

TECHNICAL FIELD

The present disclosure relates to the field of electronics component handling, in particular to the mounting of an individual electronics component at a particular placement location on a carrier.

BACKGROUND OF THE DISCLOSURE

Conventional semiconductor chip manufacturing involves the processing of semiconductor wafers with on one side an array of semiconductor devices. These wafers are then sliced into individual chips or dies using known dicing techniques after being mounted on flexible, adhesive wafer handling tape or film. The flexible, adhesive film holds the plurality of individual semiconductor dies and aids in transport during the subsequent semiconductor die handling and mounting process steps.

The individual semiconductor dies are separated from the flexible, adhesive tape by a separating apparatus, and the separated semiconductor dies are subsequently handled separated by pick-and-place equipment, e.g. placed on a lead frame, substrate, etc.

The separation of an individual semiconductor die from the flexible, adhesive film and its subsequent handling and mounting on a lead frame or substrate is susceptible to die-crack. Die-crack, scratches or chip outs are well-known failure modes in thin chip die bonding / mounting applications. Wafers with a thickness smaller than 70 µm run a high risk of die crack occurrences under mechanical or thermal stress from the handling or the processing. Because the wafer thickness relates to the product power performance, a thinner wafer improves the electrical resistance of the chip (RDSon) and minimizes the power loss.

Accordingly, in view of the need for semiconductor chips in MOSFET, SiC and GaN power applications having an even more thinner wafer thickness, it is a goal of the present disclosure to provide an improved method for mounting an electronics component at increased processing speeds at a particular placement location on a carrier as well as an improved pick-and-place apparatus for mounting an electronics component at a particular placement location on a carrier.

It is noted, that the disclosure is not limited to the mounting of various types of semiconductor dies on a carrier, but is also applicable for any similar process of handling a large number of individual electronics components to be mounted at a particular location on a carrier, such as capacitors, resistors, LEDs, etc. etc.

SUMMARY OF THE DISCLOSURE

According to a first example of the disclosure, a method for mounting an electronics component at a particular placement location on a carrier is suggested. The method comprising the steps of:

i) providing the carrier;

ii) applying an adhesive layer at least on the particular location on the carrier;

iii) positioning, using a component pick up unit holding the electronics component in an engaging manner, the electronics component above the particular placement location on the carrier;

iv) positioning, using the component pick up unit, the electronics component at a placement level above the particular placement location without contacting the adhesive layer or the carrier;

v) disengaging the component pick up unit holding the electronics component to be mounted, thereby

vi) dropping the electronics component to be mounted at the particular placement location on the carrier.

As the electronics component is dropped from a small height onto the desired location on the carrier, mechanical stresses induced on the electronics component during pick and place processes in die bond applications are minimized. This will reduce the chance of die damage or failure.

In a most simple variant of the method according to the disclosure, step vi) of dropping the electronics component is performed under the influence of earth gravity.

In a more controlled example of the method according to the disclosure, the component pick up unit holding the electronics component is constructed or configured as a vacuum die collet tool, which holds or engages the electronics component by means of a vacuum. Accordingly, step v) involves the step of releasing the vacuum of the vacuum die collet tool holding the electronics component to be mounted.

In the latter example implementing a vacuum die collet tool step vi) comprising the step of vi-1) generating and jetting, using the vacuum die collet tool, an air pulse towards (on) the electronics component to be mounted. Furthermore, the step vi-1) of jetting the air pulse is performed with an air pressure between 0.5 - 2 bar. Herewith an efficient mounting of the electronics component at a considerable short process time of 10-20 msec can be achieved.

In a further example of the method, the component pick up unit holding the electronics component comprises an engagement surface structured to engage the electronics component using a liquid pinning interface and wherein step v) involves the step of evaporating the liquid pinning interface through the supply of heat. In particular the step of evaporating the liquid pinning interface through heat comprises the step of irradiating the liquid pinning interface with laser light.

In an alternative method according to the disclosure, the component pick up unit comprises a light transparent substrate provided with a release adhesive layer engaging the at least one electronics component and wherein step v) involves the step of irradiating the release adhesive layer with laser light.

In these examples, the process time can be reduced to the order of 1-2 msec or lower.

In a further advantageous example of the method according to the disclosure, step iv) comprises the step of

setting the placement level between the electronics component and the adhesive layer on the particular placement location between 10 - 200 μm.

The disclosure also pertains to a pick-and-place apparatus for mounting an electronics component at a particular placement location on a carrier. According to the disclosure, the apparatus comprises at least:

a component pick up unit structured to pick up, hold in an engaging manner and transfer an electronics component from a pick-up location to the placement location;

a positioning unit structured to position the component pick up unit from the pick-up location to the placement location; as well as

a control unit for operating at least the component pick up unit and the positioning unit, wherein

the control unit is structured, in a first operational condition, to position the component pick up unit at a placement level above the particular placement location without contacting the substrate, and

in a second operational condition, to disengage the component pick up unit holding the electronics component to be mounted allowing the electronics component to drop at the particular placement location on the carrier.

By dropping the electronics component from a small height onto the desired location on the carrier, mechanical stresses induced on the electronics components during pick and place processes in die bond applications are minimized. This will reduce the chance of die damage or failure.

In a particular example, the component pick up unit is configured as a vacuum die collet tool by applying a vacuum on the electronics component, and wherein, in the second operational condition, the control unit is structured to release the vacuum of the vacuum die collet tool holding the electronics component. Preferably, the control unit is structured, in the second operational condition, to actuate the vacuum die collet tool in generating and jetting an air pulse towards the electronics component to be mounted.

To achieve an efficient mounting of the electronics component at a considerable short process time of 10-20 msec, the air pulse with an air pressure between 0.5 – 2 bar.

In an alternative example of the pick-and-place apparatus according to disclosure 9, the component pick up unit holding the electronics component comprises an engagement surface structured to engage the electronics component using a liquid pinning interface, and wherein, in the second operational condition, the control unit is structured to release the engagement surface holding the electronics component through the supply of heat.

In another particular example, the component pick up unit comprises a substrate provided with an release adhesive layer engaging the at least one electronics component and wherein, in the second operational condition, the control unit is structured to remove the release adhesive layer engaging the electronics component through the supply of heat.

For both example outlined above, for releasing the engagement surface engaging the electronics component or for removing the release adhesive layer engaging the electronics component the component pick up unit comprises a heating unit or a laser light device.

In a further preferred example of the pick-and-place apparatus according to the disclosure the control unit is structured to set the placement level between the electronics component and the particular placement location between 10 - 200 μm.

BRIEF DESCRIPTION OF THE DRAWINGS

The disclosure will now be discussed with reference to the drawings, which show in:

Figures 1A-1B an example of a pick-and-place apparatus according to the state of the art, depicting known method steps of mounting an electronics component at particular location on a carrier;

Figures 2A-2B a first example of a pick-and-place apparatus according to the disclosure;

Figures 3A-3C a second example of a pick-and-place apparatus according to the disclosure;

Figures 4A-4B a third example of a pick-and-place apparatus according to the disclosure;

Figures 5A-5B a fourth example of a pick-and-place apparatus according to the disclosure;

Figures 6A-6B a fifth example of a pick-and-place apparatus according to the disclosure.

DETAILED DESCRIPTION OF THE DISCLOSURE

For a proper understanding of the disclosure, in the detailed description below corresponding elements or parts of the disclosure will be denoted with identical reference numerals in the drawings.

As outlined in the introduction, in pick-and-place processing large number of individual electronics components arranged on a flexible, adhesive film, such as capacitors, resistors, LEDs, etc. etc. are to be separated from the flexible, adhesive film in an automated manner and are subsequently handled separated by pick-and-place equipment, e.g. placed on a lead frame, substrate, etc.

It is noted, that the disclosure is not limited to mounting semiconductor dies on a carrier, but is also applicable for any similar process of handling a large number of individual electronics components to be mounted at a particular location on a carrier, such as capacitors, resistors, LEDs, etc. etc.

Figures 1A-1B depict a known pick-and-place apparatus 100 according to the state of the art, depicting known method steps of mounting an electronics component 11, such as a semiconductor die or a resistor at particular location on a carrier 50. The apparatus 100 comprises at least a component pick up unit 120 structured to pick up, hold in an engaging manner and transfer an electronics component 11 from a pick-up location (not shown) to the placement location on the carrier 50. The component pick up unit 120 is provided with a pick up / engagement part 120a which engages with the electronics component 11 during the sequence of the pick up, holding, transfer and mounting steps.

The electronics component 11 (semiconductors die) has a first, top component side 11a and a second, bottom component side 11b, opposite to the first component side 11a. For a proper understanding of the disclosure, it is noted that the electronics component 11 is engaged with its first component side 11a with the pick up / engagement part 120a of the component pick up unit 120, whereas the electronics component 11 is to be mounted with its second component side 11b on the top carrier surface 50a of the carrier 50.

The phrase “engaging” as in "the electronics component 11 being engaged with the pick up / engagement part 120a” suggests that the electronics component is interacting with or making contact with the engagement part of the component pick-up unit 120. This engagement implies some form of connection or operation between both parts, such as gripping, grasping, holding, etc. or otherwise suggesting an interaction between both parts.

In a known manner, the component pick up unit 120 can be positioned by a positioning unit (not shown) from the pick-up location to the placement location above the carrier 50. Accordingly, a control unit 140 is implemented for operating at least the component pick up unit 120 (via a control line 120z) and the positioning unit.

As shown in Figure 1A, a droplet of solder 60 is provided on the top carrier surface 50a at the particular location at which the electronics component is to be mounted on the carrier 50. By means of the control unit 140, suitable control signals are send via the control line 120z based on which the component pick-up unit 120 is displaced towards the carrier 50, such that the electronics component 11 is gently pushed with its second component side 11b into the solder droplet 60. The solder droplet 60 is pushed aside and due to its liquid, viscous nature it forms a small contact layer between the bottom component side 11b and the top carrier surface 50a. This pushed mounting on the carrier 50 is susceptible to die-crack.

Figures 2A and 2B provide a first example of a pick-and-place apparatus according to the disclosure, which is capable of obviating the adverse die-crack. The first example of the pick-and-place apparatus is denoted with reference numeral 1000_1 and is capable of mounting an electronics component 11 at a particular placement location on the top carrier surface of the carrier 50.

In all examples of the pick-and-place apparatus according to the disclosure, the electronics component 11 is to be mounted on the top carrier surface 50a of the carrier 50 by means of an adhesive layer 70 which has a more or less uniform thickness and covers the particular placement location on the top carrier surface completely. In the present disclosure an interconnect paste or solder paste 60 being positioned as a droplet on the carrier 50 as in Figures 1A-1B is not used. The adhesive layer 70 can be applied on the top carrier surface 50a and at least on the particular location on the carrier 50 using various deposition techniques, such as screen printing or stencil printing. Alternatively, the adhesive layer 70 can be applied on the top carrier surface 50a using a LIFT technique (Laser Induced Forward Transfer) of an adhesive or e.g. impulse printing.

It should be noted that in the present disclosure the adhesive layer 70 being used can be formulated from a variety of materials. E.g. a solder-glue, glue, a conductive epoxy, a flux, etc. are equally suitable to be used as the adhesive layer 70. Basically any material capable connecting the carrier 50 and the electronics component 11 that can be deposited. In particular, the adhesive layer 70 is formed as a structured layer, e.g. thin film of a specific height, shape or provided with a variety of patterns.

Similarly, the pick-and-place apparatus 1000_1 is provided with a component pick up unit 120_1. The component pick-up unit 120_1 is provided with an engagement part 120a which is structured to engage with the electronics component 11 from a pick-up location to the placement location. In this example of Figures 2A-2B the engagement part 120a is capable in picking up, holding and transferring the electronics component 11. Similarly, the component pick up unit 120_1 can be positioned by a positioning unit (not shown) from the pick-up location to the placement location above the carrier 50. Accordingly, a control unit 140 is implemented for operating at least the component pick up unit 120_1 (via a control line 120z) and the positioning unit.

According to the disclosure, the control unit 140 operates differently from the prior art example. In particular, the control unit 140 is structured, in a first operational condition as depicted in Figure 2A, to position the component pick up unit 120_1 and the electronics component 11 (in particular the bottom component side 11b) at a placement level Δh above the particular placement location above the top carrier surface 50a without contacting the carrier 50. In particular, the placement level Δh between the electronics component 11 and the particular placement location on the carrier 50 is between 10 - 200 in μm.

In a second operational condition, control unit 140 disengages the component pick up unit 120_1 holding the electronics component 11 to be mounted allowing the electronics component 11 to drop at the particular placement location on the carrier 50. In the examples of the disclosures it is noted that the engagement part 120a of the component pick-up unit 120_1 can be controlled by the control unit 140 by means of the signal control line 121z. In particular, the engagement part 120a is activated when the component pick-up unit 120_1 is picking up am electronics component 11 at a pick up location. The activation (or engagement) is maintained during the subsequent lifting, holding and transferring of the electronics component 11 towards the particular placement location.

In the second operational condition, control unit 140 disengages the engagement part 120a of the component pick up unit 120_1, allowing the electronics component 11 to come loose from the component pick-up unit 120_1. This disengagement is depicted in Figure 2B by means of the large cross X placed over the signal control line 121z connecting the engagement part 120a with the control unit 140.

In the example of Figure 2B, the electronics component 11 simply drops over a distance equal to the placement level Δh under the influence of earth gravity on to the adhesive layer 70. By dropping the electronics component 11 from a small height Δh onto the desired location on the carrier 50, mechanical stresses induced on the electronics component 11 during pick and place processes in die bond applications are minimized.

This will reduce the chance of die damage or failure. In particular, in the prior art the electronics component is subjected to external forces due to being pushed into the solder paste 60 (see Figure 1B). In the examples of the present disclosure, the electronics component is subjected to inertial forces only due to its dropping, with a release velocity of about 0.1 m/s, and its mass (Impulse = moment: F *dt = m*v), resulting in impact force of ~50 micro-Newton. This is around 10000x less force applied during the know techniques of die bonding, compared to normal bonding with an attach force of 0.5 N (50 grams).

In second example of the disclosure, as shown in Figures 3A-3C, the component pick up unit 120_2 of the pick-and-place apparatus 1000_2 is configured as a vacuum die collet tool. The engagement part 120a of this example is configured as a vacuum unit 121 which engages, in the first operational condition and prior to that, the electronics component 11 at the engagement part 120a by applying a vacuum (under pressure) on the top component side 11a of the electronics component 11. This vacuum is depicted in Figure 3A by means of upward pointing arrow P implying an air suction action.

In the second operational condition, see Figure 3B, the control unit 140 releases the vacuum of the vacuum unit 121 of the vacuum die collet tool 120_2 holding the electronics component 11. This release or removal of vacuum (or under pressure) is depicted in Figure 3B by means of a large cross X through upward pointing arrow P. In a similar fashion, the release or disengagement of the vacuum unit 121, hence the removal of the suction between the engagement part 120a and the electronics component 11 causes the latter to drop over a distance equal to the placement level Δh under the influence of earth gravity on to the adhesive layer 70.

In an alternative example, shown in Figure 3C, the control unit 140 may, in the second operational condition, actuate – via the signal control line 121z - the vacuum unit 121 of the vacuum die collet tool 120_2 in generating and jetting or exerting an air pulse towards (on) the top component side 11a of the electronics component 11. This is shown in Figure 3C as the downward pointed arrow P implying an air pulse jetting action. To achieve an efficient mounting of the electronics component 11 at a considerable short process time of 10-20 msec, the air pulse is jetted with an air pressure between 0.5 – 2 bar.

In a third example of the disclosure, as shown in Figures 4A-4BC, the component pick up unit 120_3 of the pick-and-place apparatus 1000_3 is configured to engage, in the first operational condition and prior to that, the top component side 11a of the electronics component 11 at the engagement part 120a using a liquid pinning interface 80. See Figure 4A.

In the second operational condition, see Figure 4B, the control unit 140 is structured to release the engagement surface 120a holding the electronics component 11 through the supply of heat. To this end, in the example of the pick-and-place apparatus 1000_3, the component pick up unit 120_3 comprises a heating unit 122, which can be controlled by the control unit 140 via the control signal 122z. The heating unit 122 has a heating element which is in heat exchanging contact with the engagement surface 120a. By supplying heat (see the heat symbol in Figure 4B) to the engagement surface 120a the liquid pinning interface 80 will evaporate, thereby disengaging the electronics component 11 from the engagement surface 120a. Similarly, this evaporation and disengagement will cause the electronics component 11 to drop over a distance equal to the placement level Δh under the influence of earth gravity on to the adhesive layer 70.

In yet another fourth example, see Figures 5A-5B, the pick-and-place apparatus 1000_4 implements a component pick up unit 120_4 which comprises a laser light device 123, which can be controlled by the control unit 140 via the signal control line 123z. This fourth example bears similarities with the third example of Figures 4A-4B. Also in this fifth example, a liquid pinning interface 80 is used to engage or hold the top component side 11a of the electronics component 11 to the engagement part 120a. In the second operational condition, the control unit 140 will activate the laser light device 123 which will emit light (denoted in Figure 5B by $h.\vartheta$). This light represents heat, and similarly will cause the liquid pinning interface 80 to evaporate. Likewise, the electronics component 11 will disengage from the engagement surface 120a, causing the electronics component 11 to drop over a distance equal to the placement level Δh under the influence of earth gravity on to the adhesive layer 70.

In a fifth example, see Figures 6A-6B, the pick-and-place apparatus 1000_5 implements a component pick up unit 120_5 which engages at the engagement part 120a with a light transparent substrate 90. The light transparent substrate 90 is provided with a release adhesive layer 91, which engages the at least one electronics component 11. Preferably the light transparent substrate 90 carries a large number of electronics components 11-11’. Similarly, the component pick up unit 120_5 comprises a laser light device 123, which can be controlled by the control unit 140 via the signal control line 123z.

In the second operational condition, the control unit 140 will activate the laser light device 123 which will emit light (denoted in Figure 6B by h.ϑ) towards the release adhesive layer 91 and the electronics component 11 to be disengaged. This light represents heat, and similarly will cause the release adhesive layer 91 to loose its adhesive strength. Likewise, the electronics component 11 will disengage from the release adhesive layer 91, causing the electronics component 11 to drop over a distance equal to the placement level Δh under the influence of earth gravity on to the adhesive layer 70.

The disclosure also relates to a method for mounting an electronics component 11 at a particular placement location on a carrier 50. The method comprising the steps of:

i) providing the carrier 50;

ii) applying an adhesive layer 70 at least on the particular location on the carrier 50 using screen printing or by stencil printing;

iii) positioning, using a component pick up unit 120 holding the electronics component 11 in an engaging manner, the electronics component 11 above the particular placement location on the carrier 50;

iv) positioning, using the component pick up unit 120, the electronics component 11 at a placement level above the particular placement location without contacting the adhesive layer 70 or the carrier 50;

v) disengaging the component pick up unit 120 holding the electronics component 11 to be mounted, thereby

vi) dropping the electronics component 11 to be mounted at the particular placement location on the carrier 50.

As the electronics component 11 is dropped from a small height onto the desired location on the carrier 50, mechanical stresses induced on the electronics component 11 during pick and place processes in die bond applications are minimized. This will reduce the chance of die damage or failure.

In a most simple variant of the method according to the disclosure, step vi) of dropping the electronics component 11 is performed under the influence of earth gravity.

In a more controlled example of the method according to the disclosure, the component pick up unit holding the electronics component 11 is constructed or configured as a vacuum die collet tool, which holds or engages the electronics component by means of a vacuum. Accordingly, step v) involves the step of releasing the vacuum of the vacuum die collet tool holding the electronics component 11 to be mounted.

In the latter example implementing a vacuum die collet tool step vi) comprising the step of vi-1) generating and jetting, using the vacuum die collet tool, an air pulse towards or on the electronics component 11 to be mounted. Furthermore, the step vi-1) of jetting the air pulse is performed with an air pressure between 0.5 - 2 bar. Herewith an efficient mounting of the electronics component 11 at a considerable short process time of 10-20 msec can be achieved.

In a further example of the method, the component pick up unit 120 holding the electronics component 11 comprises an engagement surface 120a structured to engage the electronics component using a liquid pinning interface 80 and wherein step v) involves the step of evaporating the liquid pinning interface 80 through the supply of heat. In particular the step of evaporating the liquid pinning interface 80 through heat comprises the step of irradiating the liquid pinning interface 80 with laser light 123.

In an alternative method according to the disclosure, the component pick up unit comprises a light transparent substrate 90 provided with a release adhesive layer 91 engaging the at least one electronics component 11-11’ and wherein step v) involves the step of irradiating the release adhesive layer 91 with laser light.

In a further advantageous example of the method according to the disclosure, step iv) comprises the step of

setting the placement level between the electronics component 11 and the adhesive layer 70 on the particular placement location between 10 - 200 µm.

All applications discussion in this disclosure have the additional advantage, next to an increased processing mounting speed of an electronics component, of a reduction in the use of glue when dropping the electronics component on the carrier. This advantage of improved sustainability lies in the fact that the glue volume being used will be less compared to the known methods, e.g. as depicted in Figures 1A-1B. An efficient printing of the bonding layer - adhesive or solder underneath the electronics component – will reduce the material consumption by a factor of 2 to even over 10x, depending on the layer thickness, ranging from 5-25 µm meter

LIST OF REFERENCE NUMERALS USED

11-11’	electronics component
11a	first, top component surface
11b	second, bottom component surface
50	carrier (lead frame, substrate)
50a	top carrier surface
60	interconnect paste
70	adhesive layer
80	liquid pinning interface
90	transparent glass or foil film for multiple electronics components
91	release adhesive layer
100	pick-and-place apparatus (state of the art)
120	component pick up unit (state of the art)
1000_n	pick-and-place apparatus (1st-2nd-3rd-4th-5th example)
120_n	component pick up unit (1st-2nd-3rd-4th-5th example)
120a	pick up / engagement part
120z	signal line for component pick up unit
121	suction means of component pick up unit
121z	control line for engagement part
122	heating unit
122z	control line for heating unit
123	laser light device
123z	control line for laser light device
140	control unit
$h.\vartheta$	light energy
Δh	dropping level

CLAIMS

1. A method for mounting an electronics component at a particular placement location on a carrier, the method comprising the steps of:

i) providing the carrier;

ii) applying an adhesive layer at least on the particular location on the carrier;

iii) positioning, using a component pick up unit holding the electronics component in an engaging manner, the electronics component above the particular placement location on the carrier;

iv) positioning, using the component pick up unit, the electronics component at a placement level above the particular placement location without contacting the adhesive layer or the carrier;

v) disengaging the component pick up unit holding the electronics component to be mounted, thereby

vi) dropping the electronics component to be mounted at the particular placement location on the carrier.

2. The method according to claim 1, wherein step vi) is performed under the influence of earth gravity.

3. The method according to claim 1 or 2, wherein the component pick up unit holding the electronics component is constructed as a vacuum die collet tool holding the electronics component by means of a vacuum and wherein step v) involves the step of releasing the vacuum of the vacuum die collet tool holding the electronics component to be mounted.

4. The method according to claim 3, wherein step vi) comprising the step of:

vi-1) generating and jetting, using the vacuum die collet tool, an air pulse towards (on) the electronics component to be mounted.

5. The method according to claim 4, wherein step vi-1) of jetting the air pulse is performed with an air pressure between 0.5 - 2 bar.

6. The method according to claim 1 or 2, wherein the component pick up unit holding the electronics component comprises an engagement surface structured to engage the electronics component using a liquid pinning interface and wherein step

v) involves the step of evaporating the liquid pinning interface through the supply of heat.

7. The method according to claim 6, wherein step of evaporating the liquid pinning interface through heat comprises the step of irradiating the liquid pinning interface with laser light.

8. The method according to claim 1 or 2, wherein the component pick up unit comprises a light transparent substrate provided with a release adhesive layer engaging the at least one electronics component and wherein step v) involves the step of irradiating the release adhesive layer with laser light.

9. The method according to any one or more of the preceding claims, wherein step iv) comprises the step of

setting the placement level between the electronics component and the adhesive layer on the particular placement location between 10 - 200 μm.

10. A pick-and-place apparatus for mounting an electronics component at a particular placement location on a carrier, the apparatus comprising at least:

a component pick up unit structured to pick up, hold in an engaging manner and transfer an electronics component from a pick-up location to the placement location;

a positioning unit structured to position the component pick up unit from the pick-up location to the placement location; as well as

a control unit for operating at least the component pick up unit and the positioning unit, wherein

the control unit is structured, in a first operational condition, to position the component pick up unit at a placement level above the particular placement location without contacting the substrate, and

in a second operational condition, to disengage the component pick up unit holding the electronics component to be mounted allowing the electronics component to drop at the particular placement location on the carrier.

11. The pick-and-place apparatus according to claim 10, wherein the component pick up unit is configured as a vacuum die collet tool by applying a vacuum on the electronics component, and wherein, in the second operational condition, the control unit is structured to release the vacuum of the vacuum die collet tool holding the electronics component.

12. The pick-and-place apparatus according to claim 11, wherein the control unit is structured, in the second operational condition, to actuate the vacuum die collet

tool in generating and jetting an air pulse towards (on) the electronics component to be mounted.

13. The pick-and-place apparatus according to claim 10, wherein the component pick up unit holding the electronics component comprises an engagement surface structured to engage the electronics component using a liquid pinning interface, and wherein, in the second operational condition, the control unit is structured to release the engagement surface holding the electronics component through the supply of heat.

14. The pick-and-place apparatus according to claim 10, wherein the component pick up unit comprises a light transparent substrate provided with an release adhesive layer engaging the at least one electronics component and wherein, in the second operational condition, the control unit is structured to remove the release adhesive layer engaging the electronics component through the supply of heat.

15. The pick-and-place apparatus according to claim 13 or 14, wherein the component pick up unit comprises a heating unit or a laser light device.

ABSTRACT

According to a first example of the disclosure, a method for mounting an electronics component at a particular placement location on a carrier is suggested. The method comprising the steps of:

i) providing the carrier;

ii) applying an adhesive layer at least on the particular location on the carrier;

iii) positioning, using a component pick up unit holding the electronics component in an engaging manner, the electronics component above the particular placement location on the carrier;

iv) positioning, using the component pick up unit, the electronics component at a placement level above the particular placement location without contacting the adhesive layer or the carrier;

v) disengaging the component pick up unit holding the electronics component to be mounted, thereby

vi) dropping the electronics component to be mounted at the particular placement location on the carrier.

Figure 2B